

Quantum-enhanced Markov chain Monte Carlo (QeMCMC) for combinatorial optimization (*QECO*)

Kate V. Marshall, Daniel J. Egger, Michael Garn (STFC), Francesca Schiavello (OQI), Sebastian Brandhofer, Christa Zoufal and Stefan Woerner

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Quantum-enhanced Markov chain Monte Carlo

- *Motivation*

David Layden,^{1,*} Guglielmo Mazzola,² Ryan V. Mishmash,¹ Mario Motta,¹ Pawel Wocjan,³ Jin-Sung Kim,^{1,†} and Sarah Sheldon¹

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Sampling from complicated probability distributions is a hard computational problem arising in many fields, including statistical physics, optimization, and machine learning. Quantum computers have recently been used to sample from complicated distributions that are hard to sample from classically, but which seldom arise in applications. Here we introduce a quantum algorithm to sample from distributions that pose a bottleneck in several applications, which we implement on a superconducting quantum processor. The algorithm performs Markov chain Monte Carlo (MCMC), a popular iterative sampling technique, to sample from the Boltzmann distribution of classical Ising models. In each step, the quantum processor explores the model in superposition to propose a random move, which is then accepted or rejected by a classical computer and returned to the quantum processor, ensuring convergence to the desired Boltzmann distribution. We find that this quantum algorithm converges in fewer iterations than common classical MCMC alternatives on relevant problem instances, both in simulations and experiments. It therefore opens a new path for quantum computers to solve useful—not merely difficult—problems in the near term.

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- *Motivation – can QeMCMC be useful for combinatorial optimization?*

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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

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- *Background*
- *Our algorithm*
- *Problem instances*
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- *Future work*

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- *Background*
 - *MCMC*

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- *Background*
 - *MCMC*

$$P(s' \mid s) = Q(s' \mid s) A(s' \mid s)$$

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- *Background*
- *MCMC*

$$P(s' | s) = Q(s' | s) A(s' | s)$$

$$\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \cdot A(s' | s)$$

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- *MCMC*

$$P(s' | s) = Q(s' | s) A(s' | s)$$

$$\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \cdot A(s' | s)$$

M-H

$$A(s' | s) = \min\left(1, \frac{\mu(s') Q(s | s')}{\mu(s) Q(s' | s)}\right)$$

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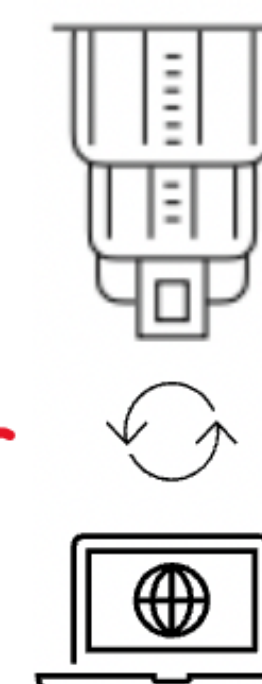
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- *Background*
 - *MCMC*
 - *QeMCMC*

$$P(s' | s) = Q(s' | s) A(s' | s)$$

$$\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \cdot A(s' | s)$$



$$|\langle s' | U | s \rangle| = |\langle s | U | s' \rangle|$$

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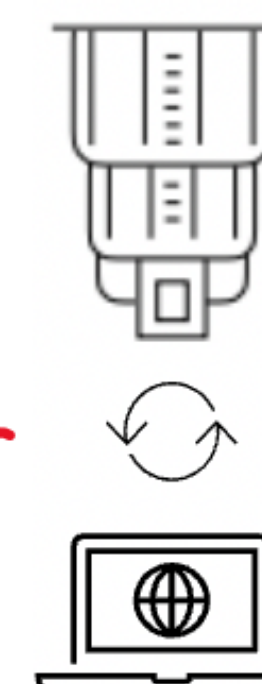


- *Background*
- *MCMC*
- *QeMCMC*

$$P(s' | s) = Q(s' | s) A(s' | s)$$

$$\begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix} \cdot \boxed{A(s' | s)}$$

$$A(s' | s) = \min \left(1, \frac{\mu(s') Q(s | s')}{\mu(s) Q(s' | s)} \right)$$



$$|\langle s' | U | s \rangle| = |\langle s | U | s' \rangle|$$

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- *Background*
 - *MCMC*
 - *QeMCMC*
 - *QeMCMC for optimization*

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- *Background*
 - *MCMC*
 - *QeMCMC*
 - *QeMCMC for optimization*
 - *Simulated annealing*
 - *Parallel tempering*

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Quantum-enhanced Markov chain Monte Carlo (QeMCMC) for combinatorial optimization (QECCO)

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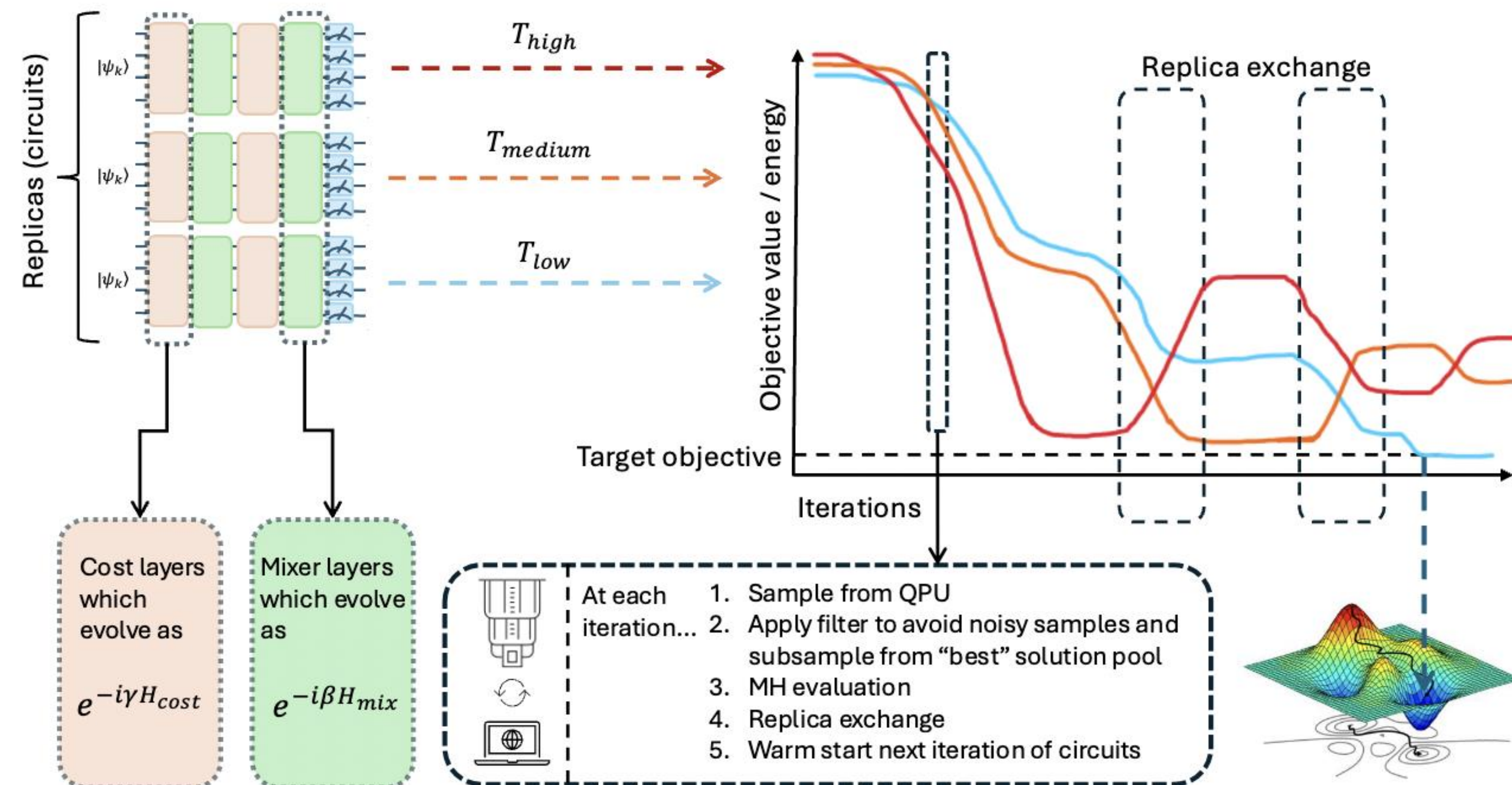
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- *Background*
 - MCMC
 - QeMCMC
 - QeMCMC for optimization
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 - Parallel tempering



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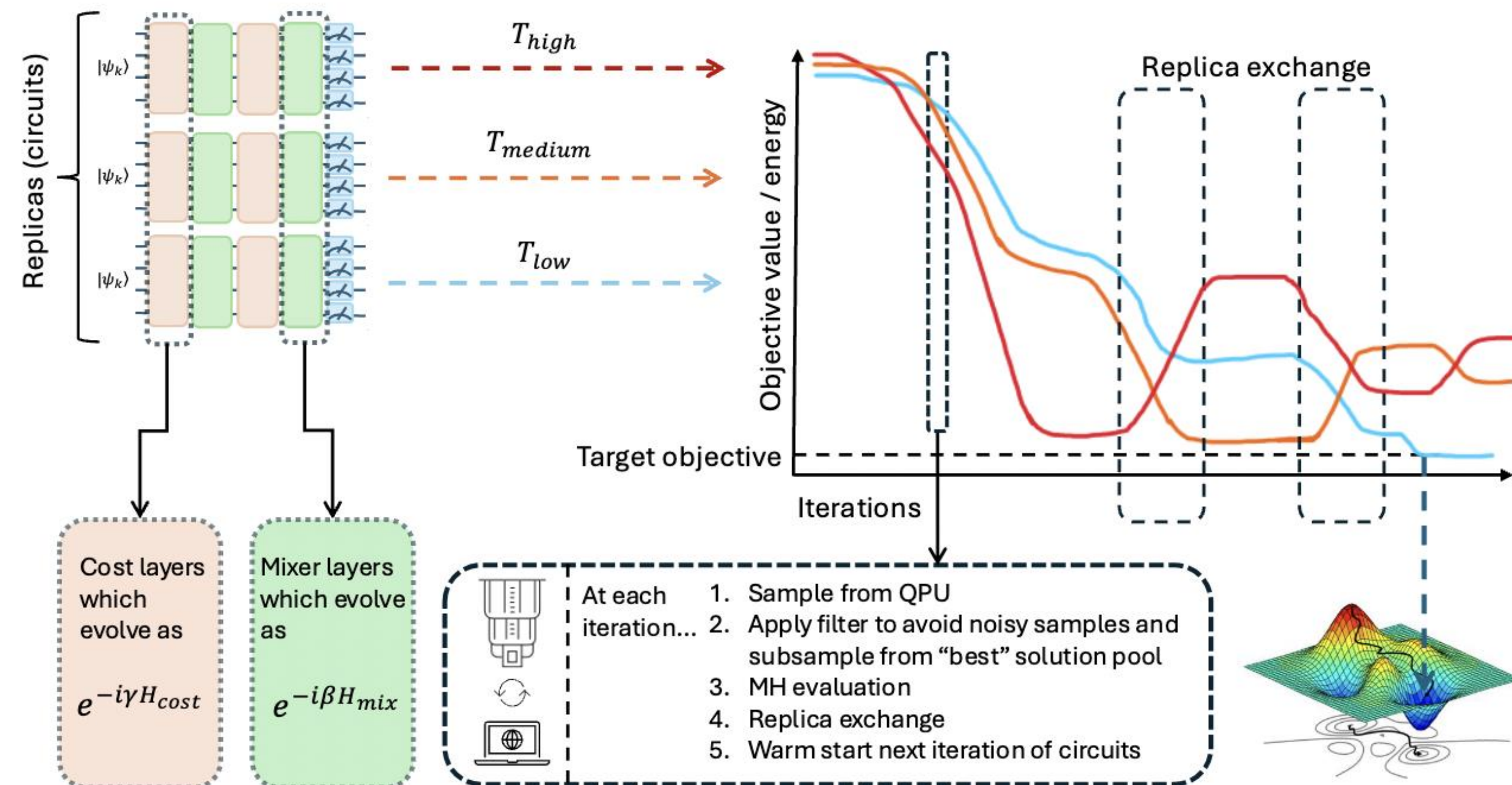
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

$$H = (1 - \kappa) \alpha H_{\text{cost}} + \kappa H_{\text{mix}}$$

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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

$$H = (1 - \kappa) \alpha H_{\text{cost}} + \kappa H_{\text{mix}}$$

We construct circuits from Eq. (10) that resemble QAOA, using a first-order Trotter decomposition, since Trotter error is not relevant. Since $[H_{\text{cost}}, H_{\text{mix}}] \neq 0$, Eq. (10) cannot be implemented exactly as a single exponential. The Lie–Trotter decomposition,

$$e^{-itH} \approx e^{-it(1-\kappa)\alpha H_{\text{cost}}} e^{-it\kappa H_{\text{mix}}} + \mathcal{O}(t^2), \quad (16)$$

divided into two identical Trotter steps yields

$$e^{-itH} \approx (e^{-i\gamma H_{\text{cost}}} e^{-i\beta H_{\text{mix}}})^2, \quad (17)$$

with fixed angles

$$\gamma = \frac{t}{2}(1 - \kappa)\alpha, \quad \beta = \frac{t}{2}\kappa. \quad (18)$$

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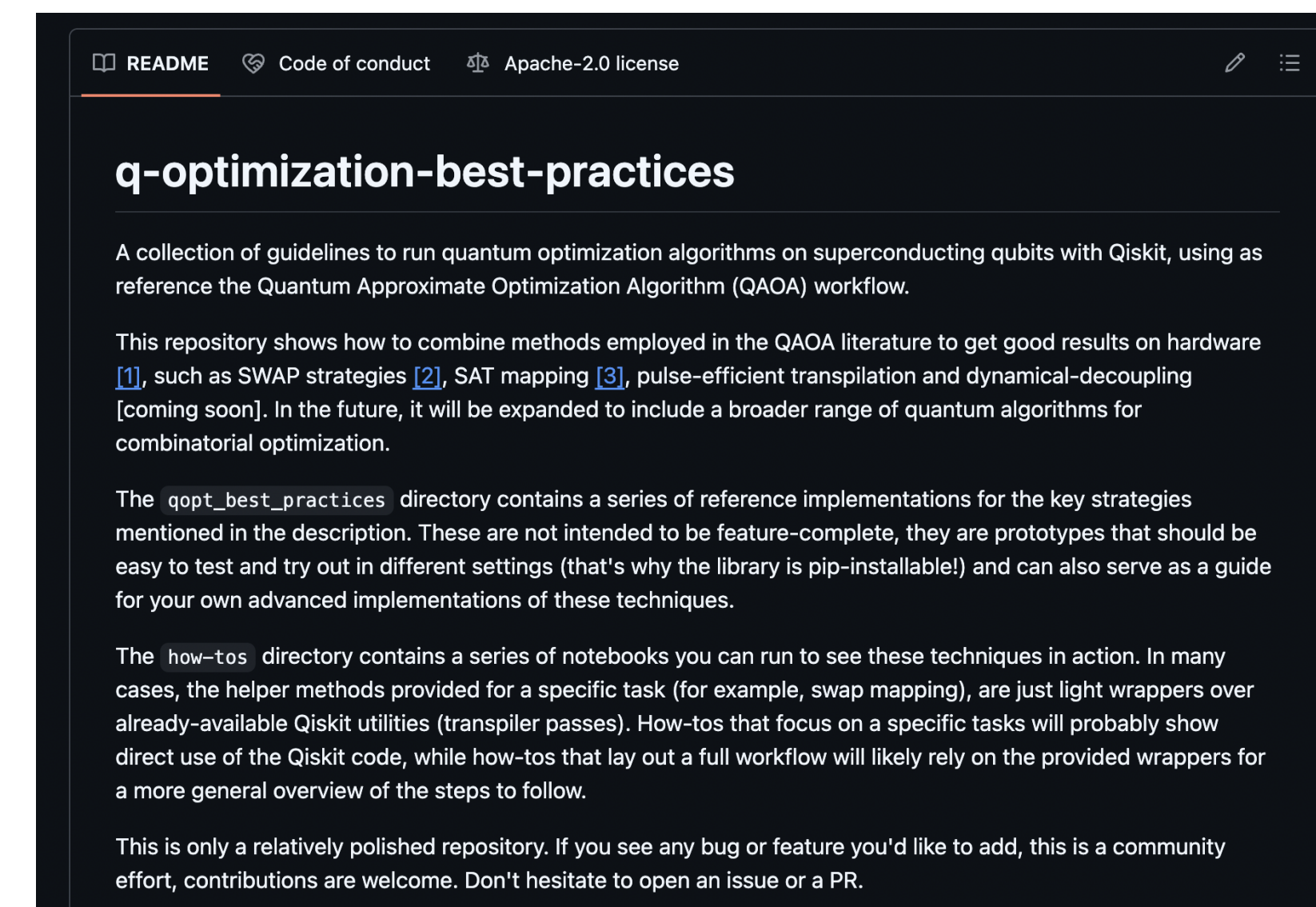
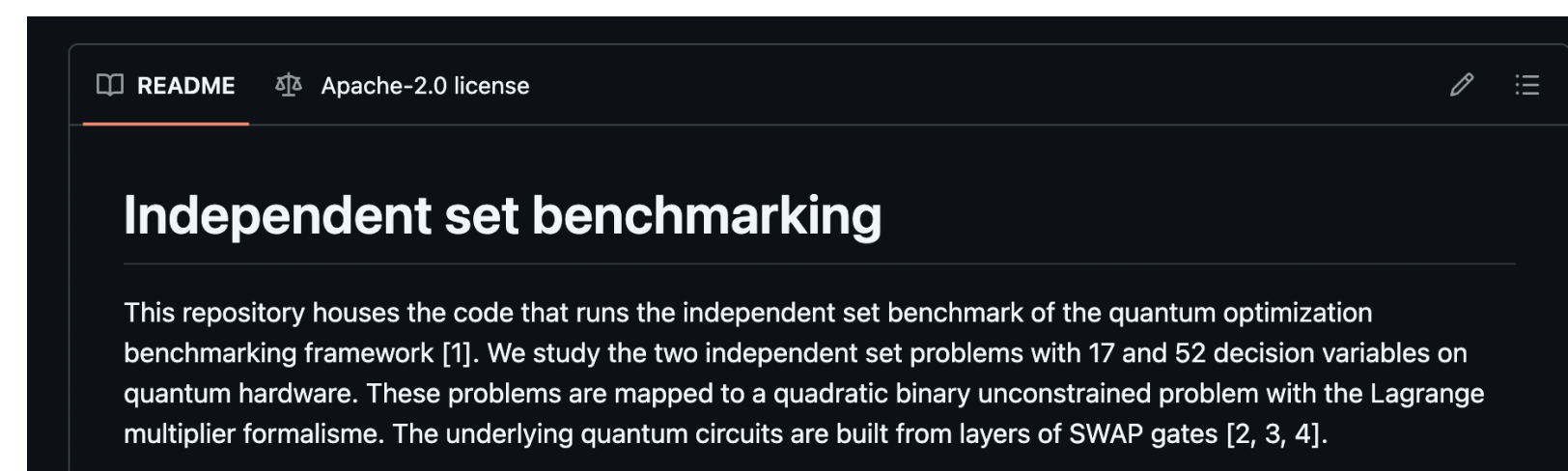
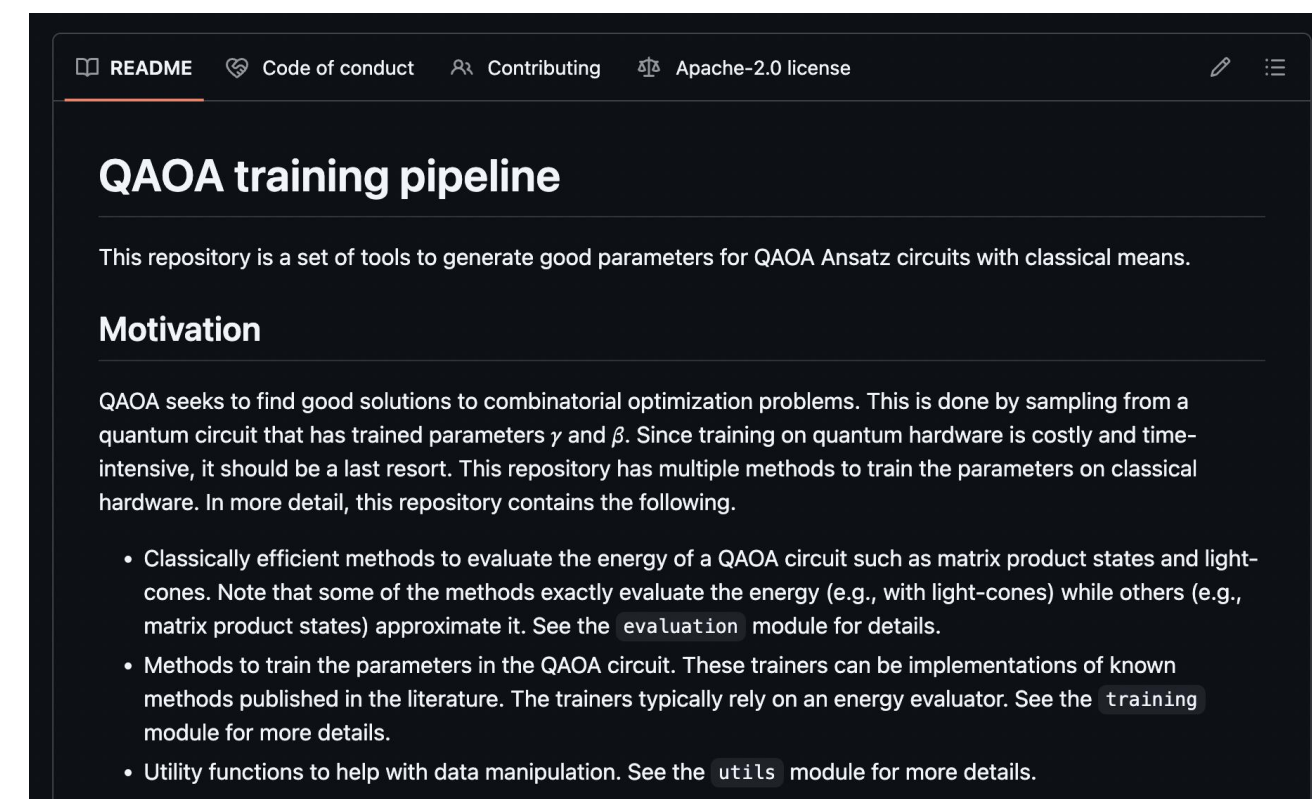
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

Warm-starting quantum optimization

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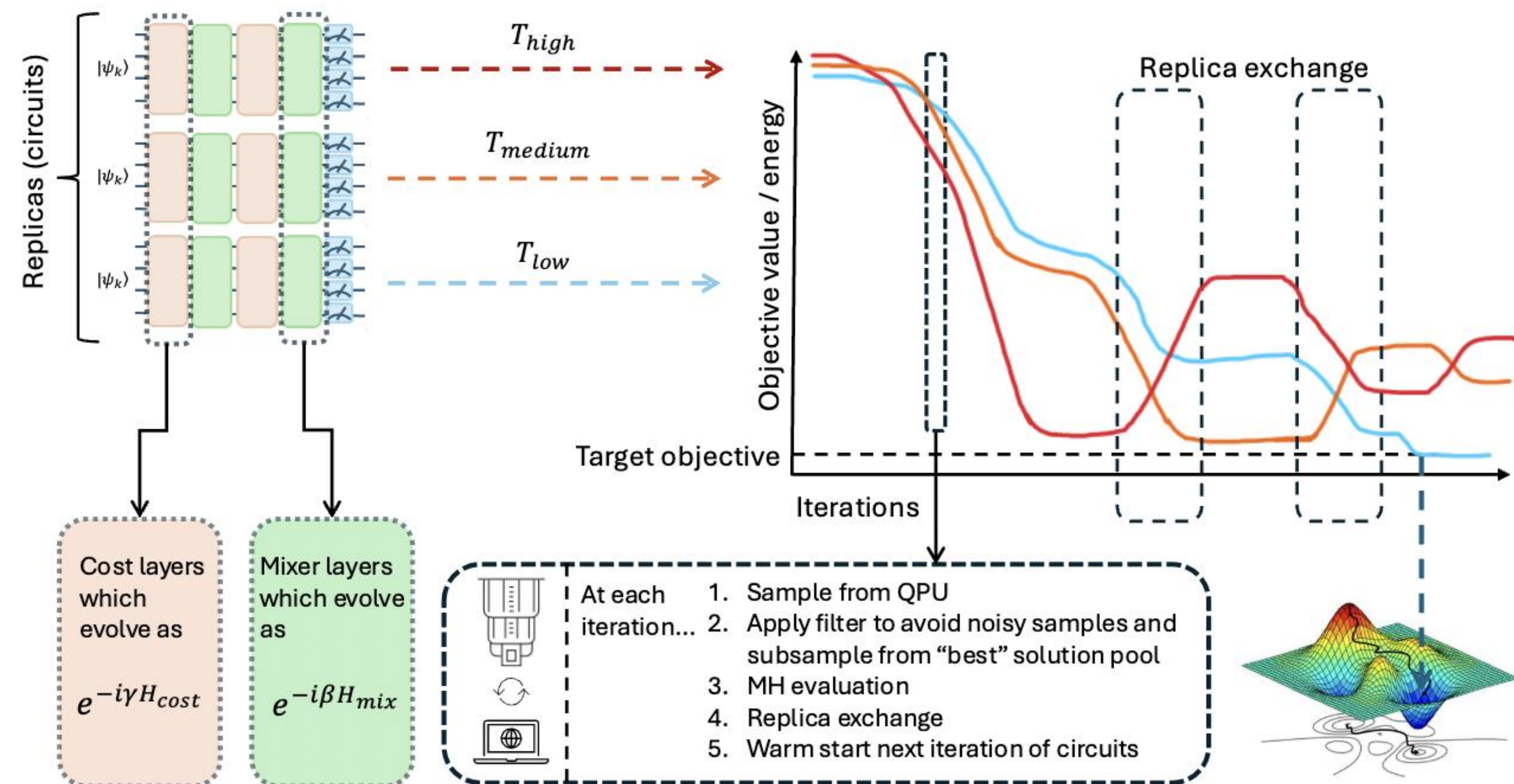
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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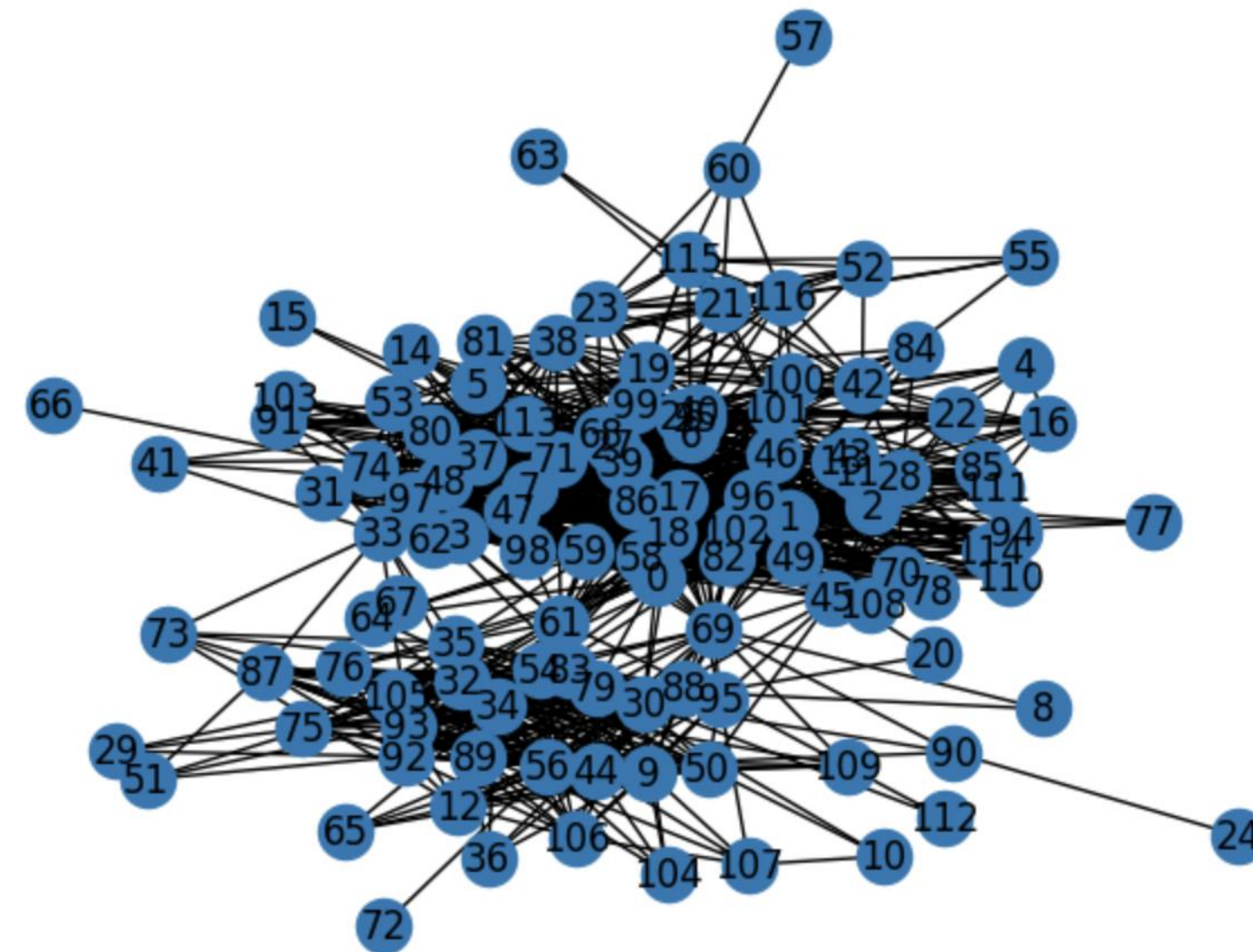
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



$$\max_x \sum_{i=0}^{|V|-1} x_i, \quad (6)$$

subject to the constraints

$$x_i x_j = 0 \quad \forall (v_i, v_j) \in E, \quad (7)$$

which enforce the selected subset of vertices to be an independent set. Further, we can formulate this as the following QUBO:

$$\max_x \left(\sum_{v_i \in V} x_i - \lambda \sum_{(v_i, v_j) \in E} x_i x_j \right), \quad (8)$$

where the first term is the objective and the second term enforces the constraints as long as λ is sufficiently large, e.g., $\lambda \geq 2$, i.e., the maximum in Eq. (8), indeed gives an optimal MIS solution.

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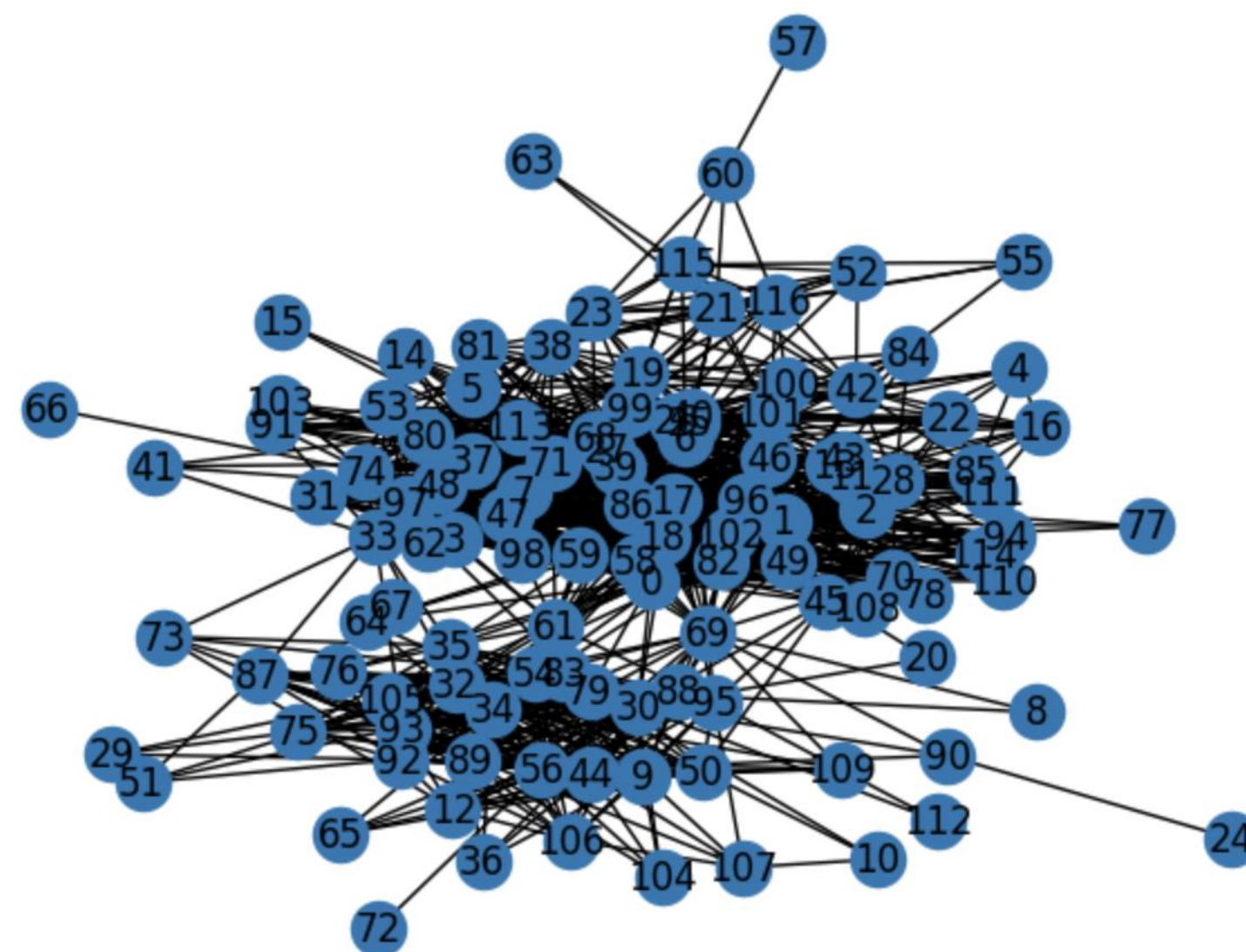
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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$$H_{\text{cost}} = H_{\text{objective}} + \lambda H_{\text{constraint}}$$

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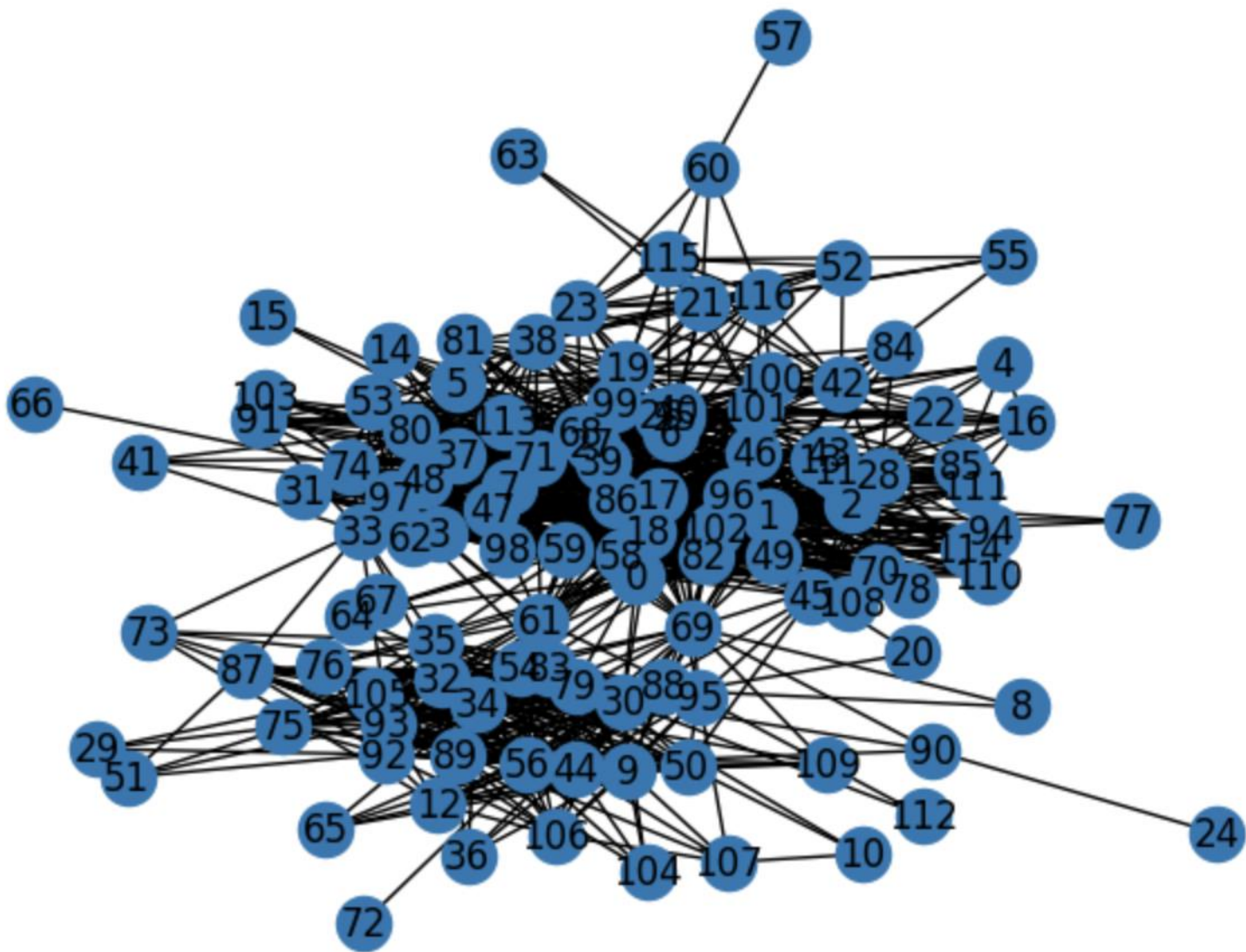
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



QOBLIB - Quantum Optimization Benchmarking Library

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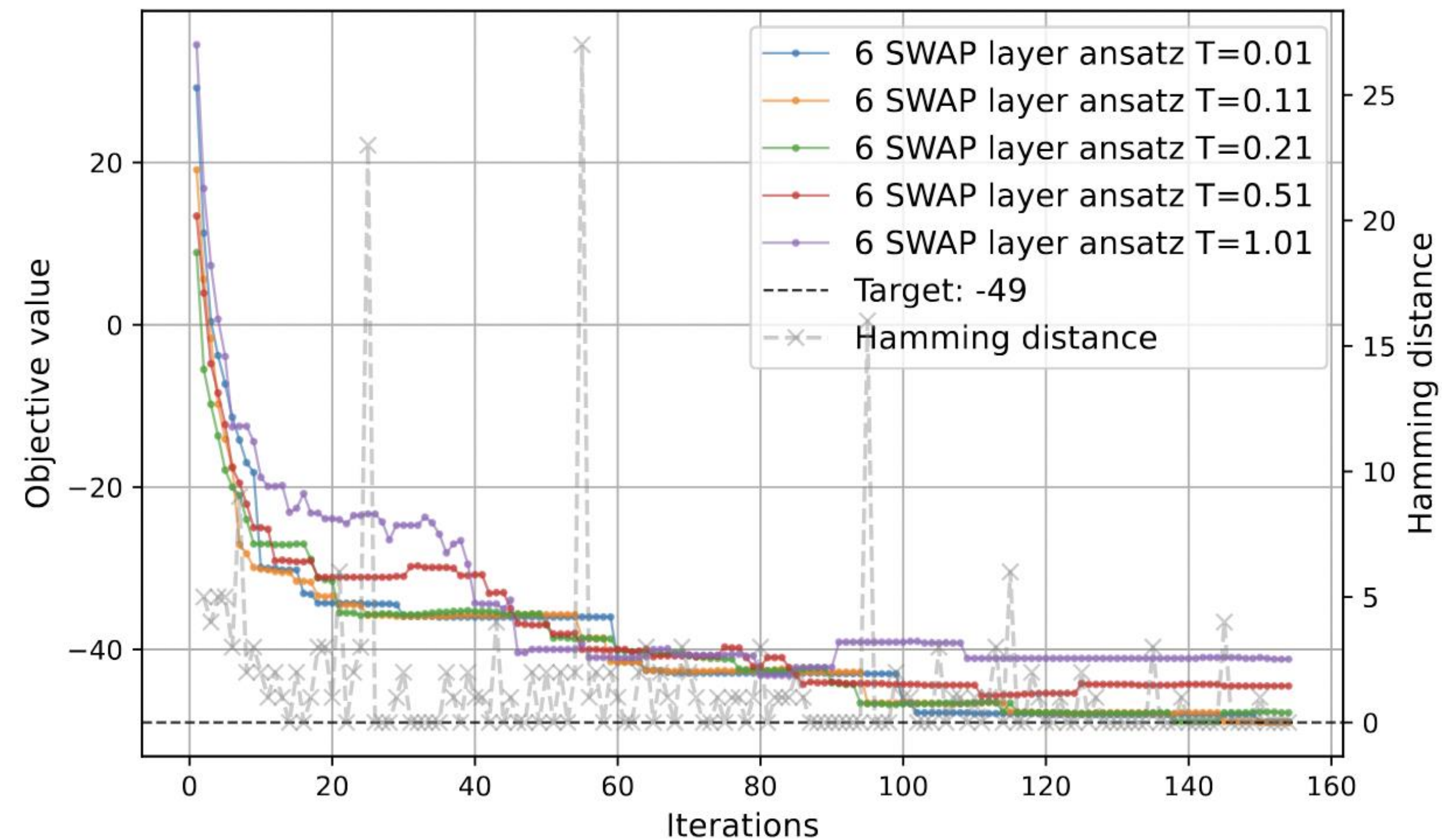
Update 1.1.0
Maximilian Schicker authored 1 month ago 18426267 History

Name	Last commit	Last update
01-marketsplit	Update 1.1.0	1 month ago
02-labs	Update 1.1.0	1 month ago
03-birkhoff	Update 1.1.0	1 month ago
04-steiner	Update 1.1.0	1 month ago
05-sports	Update 1.1.0	1 month ago
06-portfolio	Update 1.1.0	1 month ago
07-independentset	Update 1.1.0	1 month ago
08-network	Update 1.1.0	1 month ago
09-routing	Update 1.1.0	1 month ago
10-topology	Update 1.1.0	1 month ago

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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

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Article

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Provable bounds for noise-free expectation values computed from noisy samples

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Check for updates

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Andreas Bärtzchi³, Stephan Eidenbenz³, Matthias Lehmkuehler⁵ &
Stefan Woerner²✉

Quantum computing has emerged as a powerful computational paradigm capable of solving problems beyond the reach of classical computers. However, today's quantum computers are noisy, posing challenges to obtaining accurate results. Here, we explore the impact of noise on quantum computing, focusing on the challenges in sampling bit strings from noisy quantum computers and the implications for optimization and machine learning. We formally quantify the sampling overhead to extract good samples from noisy quantum computers and relate it to the layer fidelity, a metric to determine the performance of noisy quantum processors. Further, we show how this allows us to use the conditional value at risk of noisy samples to determine provable bounds on noise-free expectation values. We discuss how to leverage these bounds for different algorithms and demonstrate our findings through experiments on real quantum computers involving up to 127 qubits. The results show strong alignment with theoretical predictions.

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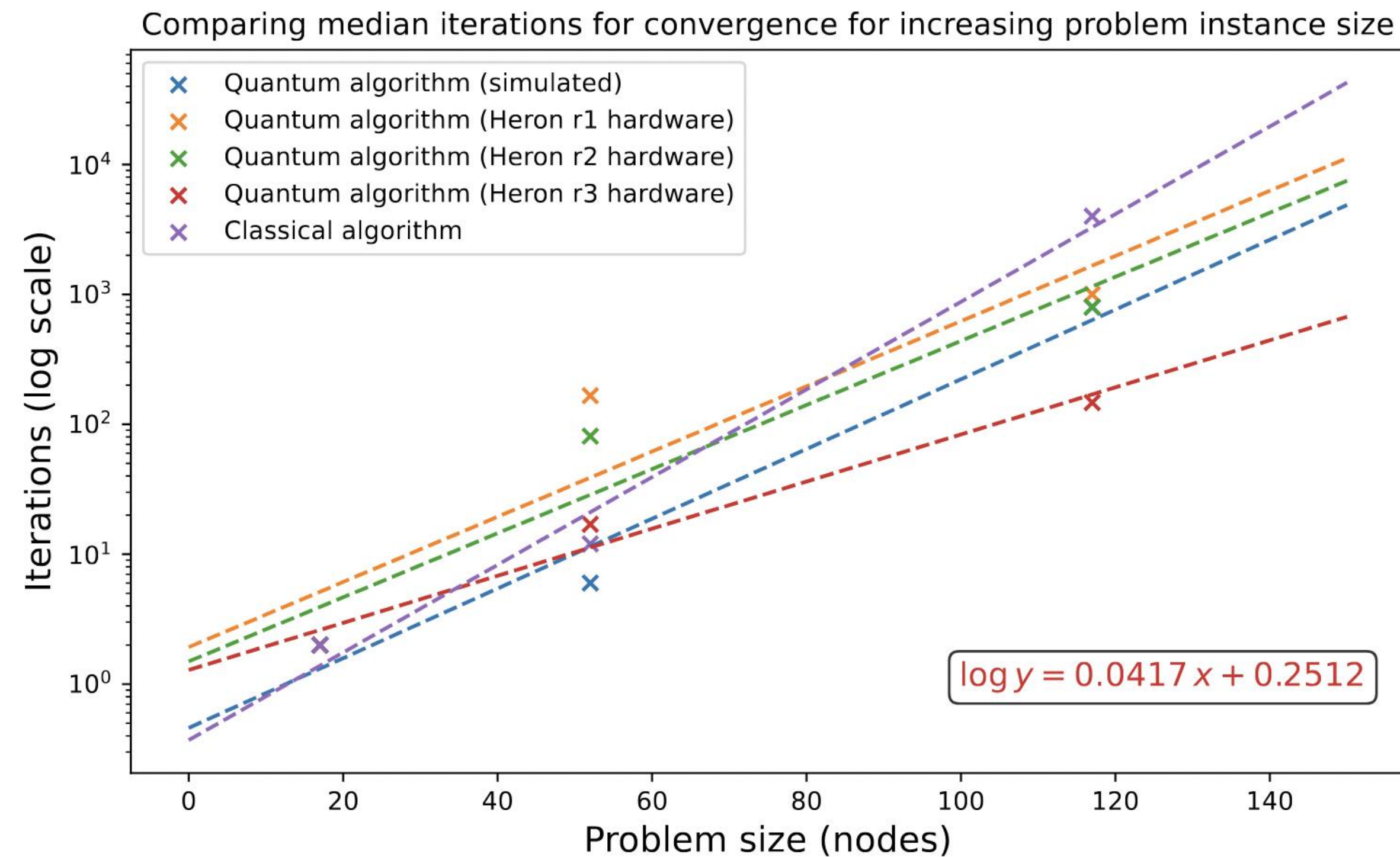
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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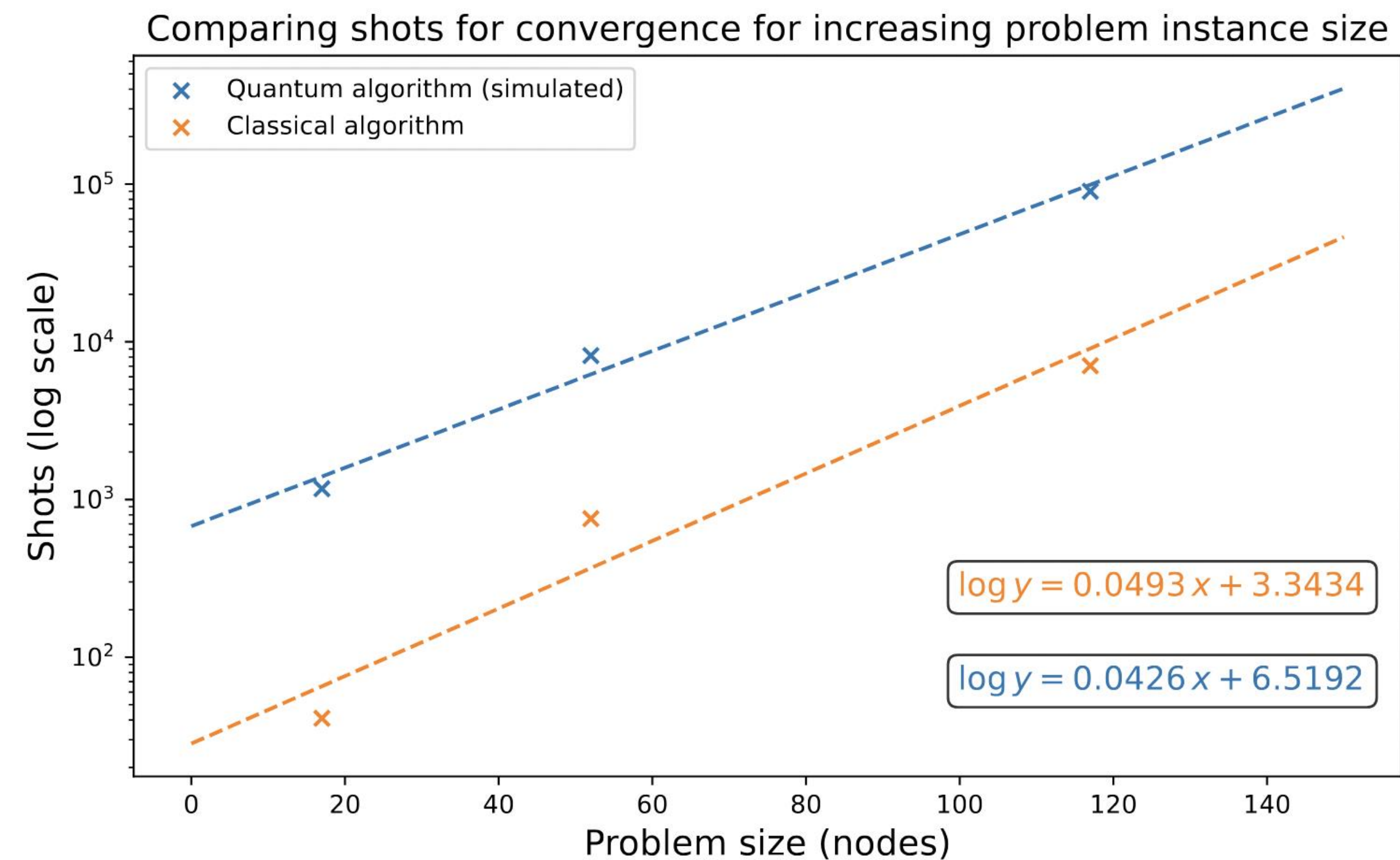
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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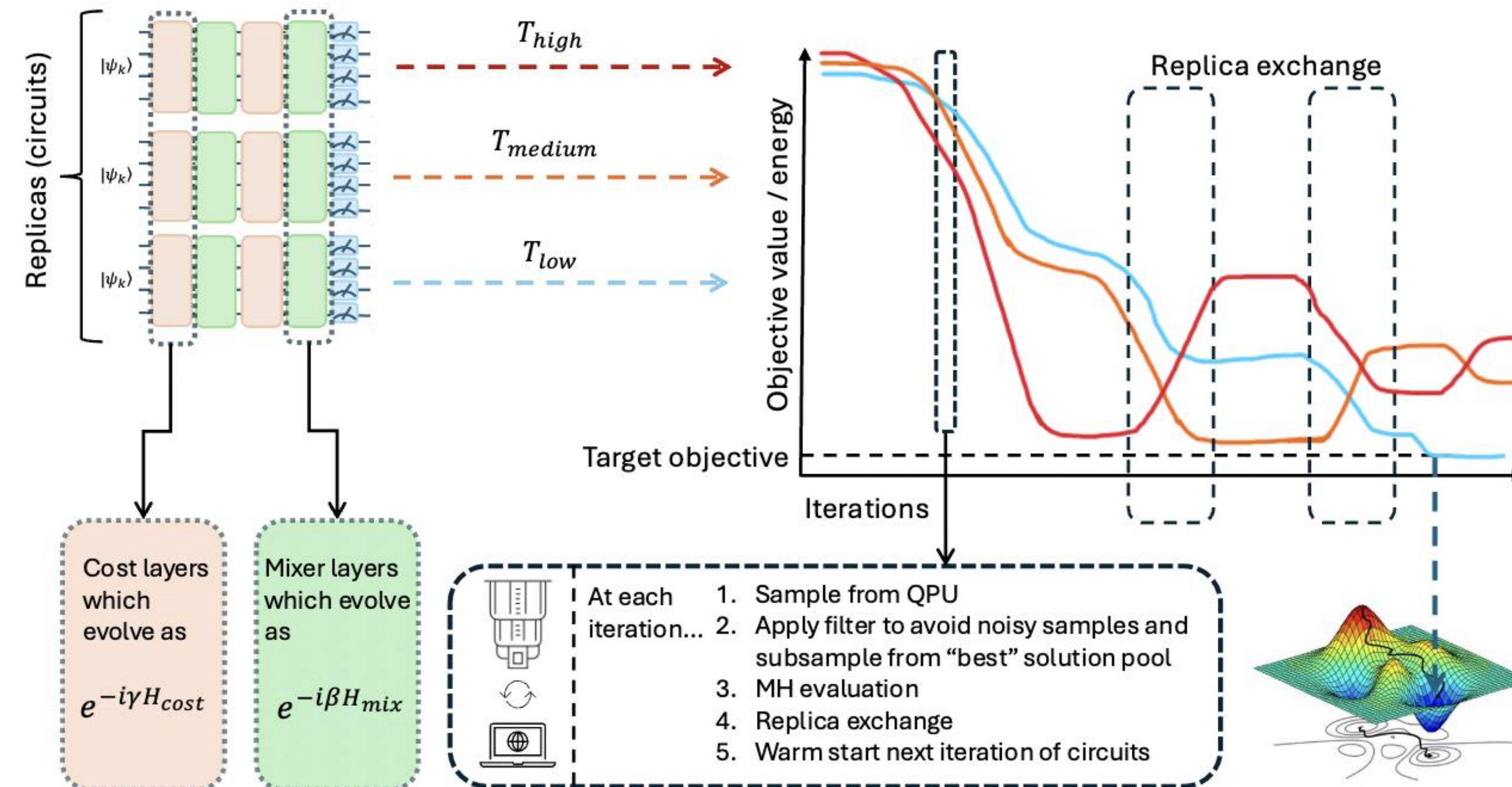
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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*



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- *Motivation*
- *Background*
- *Our algorithm*
- *Problem instances*
- *Results*
- *Future work*

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